Fiscal Year:	FY 2024	**Task Last Updated:**	FY 09/01/2023
PI Name:	Yang, Jiseon Ph.D.		
Project Title:	Microbial Social Behavior and Heritable Genetic or Epigenetic Changes Affected by the Spaceflight Environment: Understanding the Evolution of Microbial Interactions During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	11/01/2019	**End Date:**	10/31/2024
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2024 per NSSC information (Ed., 9/15/23) NOTE: End date changed to 10/31/2023 per F. Hernandez/ARC (Ed., 2/27/23)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1597		
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Task Description:	September 2023 Update: The Principal Investigator (PI) received official approval from Space Biology on May 30, 2023, to transition the Micro-17 experiment into a ground study (Ed., 9/15/23). Microbes naturally exist as mixed-species communities where they actively communicate and interact with neighboring cells. They adapt and evolve over time within the mixed-species populations. While many studies have shown that the spaceflight environment (and ground-based spaceflight analogs) impacts microbial biological processes, most of the previous experiments were based on single-species cultures. It is still unclear how the spaceflight environment influences microbial social behaviors and how their interaction in the mixed-species community contributes to microbial evolutionary processes in the spaceflight environment. Understanding the effects of the spaceflight environment on the changes in microbial social behavior and the associating evolutionary mechanisms will be necessary for spaceflight design, operations, and astronaut health. It is crucial in regard to preparation for long-duration life support systems during future exploration missions far from Earth. Microbes within the ISS are continuously exposed and adapted to the unique spaceflight environment, which includes microgravity, low fluid shear, and enhanced radiation. A variety of microbial isolates have been recovered from the ISS, including the potable water system. The ISS Water Recovery System recycles urine, condensation, and wastewater. Astronauts rely on the recycled potable water for survival. Most of the microbes recovered from ISS portable water are known biofilm formers, which can increase risks for crew infections or damage to life support system function and integrity. Therefore, it is critical to understand how microbes in the ISS water system adapt and evolve over time. The purpose of the study is to understand the microbial evolution associated with interspecies interaction and to examine the influence of the spaceflight environment on microbial interactions and evolutions in the mixed-species population, by cohort profiling. The hypothesis is that changes in microbial interaction between species of ISS water isolates are associated with microbial genomic evolution during spaceflight, and spaceflight is a contributing factor for the evolution of microbial interaction between different species.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is searching for an understanding by which microorganisms interact and evolve in the context of the complex microbial ecosystem. While immediately applicable to the unique spaceflight environment, it will also be applicable to closed environments in which water can be recycled and reused in terrestrial environments. Overall, this contributes to the creation of healthier and sustainable human built and living environments both in space and on Earth. The study uses the International Space Station as a microbial observatory to study the complex microbial ecosystems and dynamics in human built and living environments, including spaceflights. The study aligns with the research goals in the Space Biology Microbiology Element to determine the effect of spaceflight on life, process, and community dynamics. Completion of this project would 1) provide genomic evidence on the potential microevolution of ISS potable water, 2) provide insight into how microorganisms in the heterogeneous population evolve over time, and 3) determine whether the spaceflight environment is a key contributing factor that influences changes in microbial interaction. The knowledge gained from this study may a) fill knowledge gaps in microbial interspecies dynamics and evolution within the mixed-species population; b) provide novel insight into how the microbial ecological success within the mixed-species population contributes to microbial evolution in human Built Environment, whether in spaceflight or on Earth; and c) provide useful knowledge to support long-duration life support system, particularly, a water recovery system in spaceflight.
Task Progress:	This is the 4th year project report (FY2024). • Restructuring: The Micro-17 project experienced contamination and hardware failures during the 3-month Science Verification Test (SVT). The test was terminated early due to the extensive hardware failure with the reservoir design and media transfers across all three fluid processing cassettes. Following the failed SVT, the NASA Ames Space Biology Control Board (ASBCB) / NASA Biological and Physical Sciences (BPS) made the decision to restructure the Micro-17 spaceflight grant to a ground-based study in November 2022. The PI re-designed the experimental plan to complete the remaining work using a ground-based spaceflight analog system. To align with the transition of Micro-17 from a spaceflight to a spaceflight analog study, the overall goals and objectives of the original proposal will be accomplished by performing multigenerational co-culture using the NASA Rotating Wall Vessel (RWV) bioreactor. Overall, there will be no changes in scope of original science, except using a ground-based spaceflight-analog cell culture system to simulate bacterial growth during spaceflight. This transition of the Micro-17 experiment into a ground study was officially approved in May 2023. • Progress after approval: After receiving approval, significant progress was made. Dr. McLean from Texas State University kindly provided the PI team with the spaceflight analog cell culture system. This system, complete with RWV bioreactors, was successfully installed and tested in the PI's lab. The team verified biofilm formation on the membrane within the RWVs (data not provided here). Additionally, the PI team optimized the experimental methods related to inoculation, culture, and media exchange. Protocols for sample collection and processing were also refined by our team. • Experiment: Two distinct bacterial strains, sourced from the International Space Station (ISS), were inoculated into the RWV ground-based spaceflight analog cell culture systems. The long-duration co-culture is currently in progress. We obtained T1 samples from the RWV bioreactors and harvested both biofilm and planktonic samples. Further incubation is currently underway.
Bibliography Type:	Description: (Last Updated: 09/02/2025)
Articles in Peer-reviewed Journals	Santomartino R, Averesch NJH, Bhuiyan M, Cockell CS, Colangelo J, Gumulya Y, Lehner B, Lopez-Ayala I, McMahon S, Mohanty A, Santa Maria SR, Urbaniak C, Volger R, Yang J, Zea L. "Toward sustainable space exploration: A roadmap for harnessing the power of microorganisms." Nat Commun. 2023 Mar 21;14(1):1391. Review. https://doi.org/10.1038/s41467-023-37070-2 ; PMID: 36944638; PMCID: PMC10030976 , Mar-2023
Articles in Peer-reviewed Journals	Justiniano YA, Goeres DM, Sandvik EL, Kjellerup BV, Sysoeva TA, Harris JS, Warnat S, McGlennen M, Foreman CM, Yang J, Li W. "Mitigation and use of biofilms in space for the benefit of human space exploration." Biofilm. 2023 Dec 1;5:100102. https://doi.org/10.1016/j.bioflm.2022.100102 ; PMID: 36660363 PMCID: PMC9843197 , Dec-2023